A HOMOGENEOUS SPACE WITH NO NONCONSTANT QUASIHARMONIC FUNCTIONS

YITZCHAK SHMALO

ABSTRACT. Björklund asked for a countable measured group (G, μ) and a subgroup H of infinite index such that the homogeneous space G/H carries no nonconstant left μ-harmonic or μ-quasiharmonic functions, bounded or not. We give such an example. The group is the group of finitely supported permutations of $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, the subgroup is the stabiliser of 0, and μ is any probability measure that charges every transposition $(0\ n)$, $n \geq 1$, and nothing else. The induced walk on $G/H \cong \mathbb{N}_0$ is a star with infinitely many leaves. The equation at a leaf k forces a quasiharmonic function with defect c to satisfy $u(k) = u(0) - c/\mu((0\ k))$, and the equation at the centre then forces $c = 0$ and u constant. The argument uses only the convergence of the single series that defines the equation at the centre.

1. INTRODUCTION

A *measured group* is a pair (G, μ) where G is a countable group and μ is a probability measure on G whose support generates G as a semigroup. Write $\check{\mu}(s) = \mu(s^{-1})$. Following [1, Sections 1 and 4.1], a complex-valued function ϕ on G is *left μ-harmonic* if

$$\int_G \phi(sg)\, d\check{\mu}(s) = \phi(g) \qquad (g \in G), \tag{1}$$

and *left μ-quasiharmonic* if there is a constant $\ell(\phi) \in \mathbb{C}$ with

$$\int_G \phi(sg)\, d\check{\mu}(s) = \phi(g) + \ell(\phi) \qquad (g \in G), \tag{2}$$

in both cases for μ-integrable ϕ. Left translation commutes with right translation, so for a subgroup $H < G$ these equations make sense for functions on the space G/H of left cosets, that is, for right H-invariant functions on G. Constant functions are harmonic, and harmonic functions are quasiharmonic with $\ell = 0$. Quasiharmonic functions are biharmonic, and biharmonic functions on groups were studied by Björklund and Hartnick [6] in connection with quasimorphisms along random walks; for bounded harmonic functions of random walks on groups and the Liouville property see Kaimanovich and Vershik [8] and the book of Woess [9].

Björklund studies harmonic and quasiharmonic functions on homogeneous spaces in [1, Appendix II]. His Problem 2 in [1, Appendix II, Section 7.1] reads: construct a countable measured group (G, μ) and an infinite index subgroup $H < G$ such that G/H admits no non-constant left (quasi-)μ-harmonic functions whatsoever. The text before it remarks that for general quotient spaces there could well be simple counterexamples. The following theorem gives one.

Theorem 1.1. *Let $G = \mathrm{Sym}_{\mathrm{fin}}(\mathbb{N}_0)$ be the group of permutations of $\mathbb{N}_0$ that move only finitely many points, let $H = \mathrm{Stab}_G(0)$, and let μ be a probability measure on G with $\mu((0\ n)) = p_n > 0$ for every $n \geq 1$ and $\sum_{n \geq 1} p_n = 1$. Then (G, μ) is a measured group, μ is symmetric, H has*

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infinite index in G, and every left μ-quasiharmonic function on G/H is constant. In particular every left μ-harmonic function on G/H is constant.

A concrete choice is $p_n = 2^{-n}$. The proof in Section 3 shows a little more: a function u on G/H satisfying the quasiharmonic equation is constant as soon as the one series defining its left side at the coset H converges; absolute convergence is not needed (Theorem 3.1). The group G is not finitely generated and μ has infinite support. Both features are used: the centre of the star has infinitely many neighbours, and this is what rules out a nonzero defect. On G itself the picture is different: every nontrivial element of G has infinitely many conjugates, so by the theorem of Frisch, Hartman, Tamuz and Vahidi Ferdowsi [7] some symmetric non-degenerate probability measure on G of finite entropy has nonconstant bounded harmonic functions on G. The problem with G finitely generated or μ finitely supported is not treated here.

2. The measured group and its quotient

For $n \geq 1$ write $\tau_n = (0\ n)$, and let μ be as in Theorem 1.1.

Lemma 2.1. *The group G is countable, $\check{\mu} = \mu$, the support of μ generates G as a semigroup, and the map $gH \mapsto g(0)$ is a bijection from G/H onto $\mathbb{N}_0$ satisfying $(sg)(0) = s(g(0))$ for $s, g \in G$. In particular $[G : H] = \infty$.*

Proof. Every element of G lies in $\mathrm{Sym}(\{0, 1, \dots, m\})$ for some m, so G is a countable union of finite groups. Each τ_n is an involution, so $\check{\mu} = \mu$. For distinct $i, j \geq 1$ one has $\tau_i \tau_j \tau_i = (i\ j)$, since conjugating $(0\ j)$ by $(0\ i)$ replaces 0 by i. Hence the semigroup generated by the τ_n contains every transposition of $\mathbb{N}_0$, and every element of G is a finite product of transpositions. The map $gH \mapsto g(0)$ is well defined because H fixes 0; it is injective because $g(0) = g'(0)$ means $g^{-1}g' \in H$; and it is onto because $\tau_n(0) = n$. The identity $(sg)(0) = s(g(0))$ is the definition of composition. □

Identify G/H with $\mathbb{N}_0$ through Lemma 2.1, and let $u\colon \mathbb{N}_0 \to \mathbb{C}$ correspond to the right H-invariant function $\tilde{u}(g) = u(g(0))$ on G. Since $\check{\mu} = \mu$ and $\tau_n^{-1} = \tau_n$, the left side of (2) for $\tilde{u}$ at g equals $(Pu)(g(0))$, where

$$(Pu)(x) = \sum_{n \geq 1} p_n\, u(\tau_n x) \qquad (x \in \mathbb{N}_0).$$

Because $\tau_n 0 = n$, while for $k \geq 1$ one has $\tau_n k = 0$ if $n = k$ and $\tau_n k = k$ otherwise,

$$(Pu)(0) = \sum_{n \geq 1} p_n\, u(n), \tag{3}$$

$$(Pu)(k) = p_k\, u(0) + (1 - p_k)\, u(k) \qquad (k \geq 1). \tag{4}$$

This is the transition operator of the walk on the infinite star with centre 0 that jumps from the centre to the leaf n with probability p_n and from the leaf k to the centre with probability p_k, staying put otherwise. The series in (4) has only two distinct values of u in it and converges absolutely for every u; the series (3) is the only one whose convergence depends on u.

3. Proof of the theorem

Theorem 3.1. *Let $u\colon \mathbb{N}_0 \to \mathbb{C}$ and $c \in \mathbb{C}$. Suppose that the series $\sum_{n \geq 1} p_n u(n)$ converges, in the order $n = 1, 2, \dots$, and that $Pu = u + c$ on $\mathbb{N}_0$. Then $c = 0$ and u is constant.*

Proof. For $k \geq 1$, equation (4) and $(Pu)(k) = u(k) + c$ give $p_k\, u(0) + (1 - p_k)\, u(k) = u(k) + c$, that is,

$$p_k\big(u(0) - u(k)\big) = c, \qquad u(k) = u(0) - \frac{c}{p_k}. \tag{5}$$

Hence the n-th term of the series in (3) is $p_n u(n) = p_n u(0) - c$. Since $\sum_n p_n = 1$, we have $p_n \to 0$, so the terms tend to $-c$. The terms of a convergent series tend to 0, so $c = 0$, and then (5) gives $u(k) = u(0)$ for every $k \geq 1$. □

Proof of Theorem 1.1. Lemma 2.1 gives the statements about G, μ and H. If ϕ is a left μ-quasiharmonic function on G/H, then $u = \phi$ viewed on $\mathbb{N}_0$ satisfies $Pu = u + \ell(\phi)$, and the left side of (2) at $g = e$ is the convergent series $\sum_n p_n u(n)$. Theorem 3.1 shows that $\ell(\phi) = 0$ and u is constant. Harmonic functions are the case $\ell = 0$. □

Remark 3.2. The value of the defect is not normalised anywhere: at the coset H the equation reads $\ell(\phi) = \sum_n p_n u(n) - u(0)$, and nonzero constants are harmonic. The conclusion is that nonconstant solutions do not exist, not that every solution vanishes.

Remark 3.3. The same proof applies to $\mathrm{Sym}_{\mathrm{fin}}(X)$ and the stabiliser of a point x_0, for any countably infinite set X, with μ charging every transposition $(x_0\ x)$, $x \neq x_0$. Positivity of every p_n is used: if $p_k = 0$, then τ_k is not in the support and $u(k)$ may be changed freely, so the support no longer generates G and nonconstant harmonic functions appear.

Computations

The file `code/checks.py` in [2] (Python 3, standard library, exact integer and rational arithmetic; its output is `code/checks_output.txt`) checks on finite truncations the identities used above: $\tau_i\tau_j\tau_i = (i\ j)$ on $\{0, \dots, N\}$ for $N \leq 8$; that positive words in $\tau_1, \dots, \tau_N$ give all $(N+1)!$ permutations of $\{0, \dots, N\}$ for $N \leq 6$; that $gH \mapsto g(0)$ is well defined, injective on cosets and equivariant for $N \leq 5$; the rows (3) and (4) on $\{0, \dots, 12\}$; and the identities (5) and $p_k u(k) = p_k u(0) - c$ for $p_k = 2^{-k}$, $k \leq 60$. The proofs above do not depend on this file.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [3] and working together inside the Brain framework [5], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [4], the Math Brain repository [3] and the Brain framework repository [5].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In September 2026 an OpenAI GPT model, running as a research task of that project, proposed the construction and the argument. In October 2026 Claude Opus 5.5

(Anthropic), working in Claude Code, checked the proof against the source, wrote the computer checks of the repository [2], and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/bjorklund-star-quotient`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [3].

References

[1] M. Björklund, *Five remarks about random walks on groups*, preprint, 2014, arXiv:1406.0763v2.
[2] Y. Shmalo, *Code for this paper*, `https://github.com/yspennstate/bjorklund-star-quotient`.
[3] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.
[4] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.
[5] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.
[6] M. Björklund and T. Hartnick, *Biharmonic functions on groups and limit theorems for quasimorphisms along random walks*, Geom. Topol. **15** (2011), 123–143.
[7] J. Frisch, Y. Hartman, O. Tamuz and P. Vahidi Ferdowsi, *Choquet–Deny groups and the infinite conjugacy class property*, Ann. of Math. (2) **190** (2019), 307–320; arXiv:1802.00751.
[8] V. A. Kaimanovich and A. M. Vershik, *Random walks on discrete groups: boundary and entropy*, Ann. Probab. **11** (1983), 457–490.
[9] W. Woess, *Random Walks on Infinite Graphs and Groups*, Cambridge University Press, Cambridge, 2000.

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel